

Lecture 1: Large Language Models & The Transformer Architecture

A Visual and High-Level Introduction

Transformer Lectures Series

Based on 3Blue1Brown Lessons

What is a GPT Model?

Generative Pre-trained Transformer

An advanced neural network architecture optimized for processing and generating sequence data.

- ▶ **Generative:** Designed to produce new sequence data (e.g., text, code, images).
- ▶ **Pre-trained:** Trained on astronomical quantities of text data before task-specific tuning.
- ▶ **Transformer:** A specific type of deep learning network introduced in 2017.

Applications: Translation, voice-to-text, text-to-image, chatbots, and more.

The Fundamental Task: Next-Token Prediction

Under the hood, all text-generating LLMs solve a simple task repeatedly:

"Given a prompt, predict the single most likely chunk of text that follows."

► **Autoregressive Generation:**

1. Accept user input.
 2. Predict the next token (as a probability distribution).
 3. Sample a token from that distribution.
 4. Append the token to the context and repeat.
- This word-by-word (or token-by-token) execution is what you see in real-time chat interfaces.

Tokenization: Translating Text to Data

Computers do not read words; they read numbers. Input text is first sliced into **tokens**.

Example Sentence

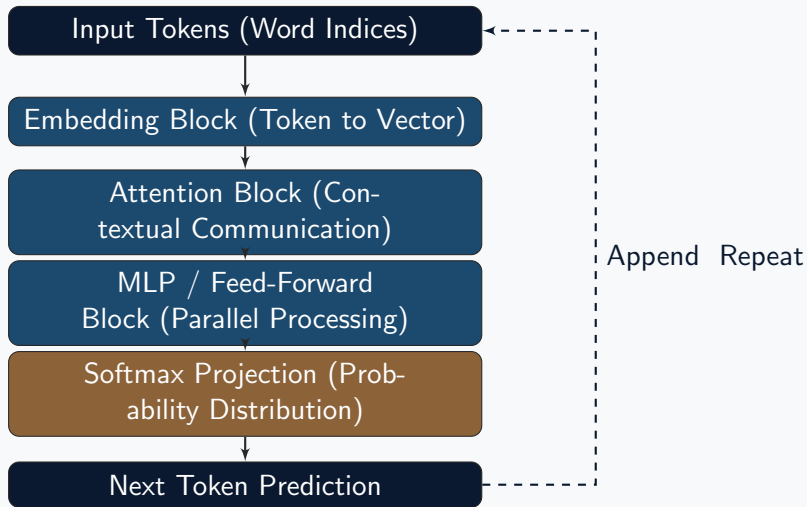
"To date, the cleverest thinker of all time was..."

- ▶ **Sub-word splitting:** Large or rare words are broken into common roots and suffixes:

[To] [date] [,] [the] [cle] [ve] [rest] [thinker] [of] [
all] [time] [was]

- ▶ **Vocabulary Map:** Every token corresponds to a specific integer ID in a fixed vocabulary (typically 50,000 to 100,000+ words/fragments).

High-Level Architecture Block Diagram



Input Embeddings: Representing Meaning Geometrically

- ▶ Once tokenized, each token is mapped to a vector in a high-dimensional space.
- ▶ **Vector Coordinates:** These represent abstract semantic features.
- ▶ **Geometric Semantics:**
 - ▶ Words with similar meanings land near each other.
 - ▶ Directions in this space encode relationships (e.g., gender, tense, country-capital).
- ▶ This vector space provides the raw material that the model will process.

The Attention Block: Contextual Communication

Static embeddings do not capture context. Attention is where words "talk" to each other.

Context matters

Compare:

- ▶ "A machine learning **model**"
 - ▶ "A famous fashion **model**"
- ▶ Attention enables the vector for **model** to communicate with **machine** or **fashion**.
 - ▶ This updates its coordinates to represent its precise contextual meaning.

The Multilayer Perceptron (MLP): Parallel Processing

After communicating via attention, the updated vectors enter an MLP (Feed-Forward) layer.

- ▶ **No Communication:** In this block, vectors do not talk to each other. They undergo the exact same mathematical operations in parallel.
- ▶ **Key Intuition:**
 - ▶ The network asks a sequence of "questions" about each token's attributes (e.g., "Is this word a verb?", "Is this noun singular?").
 - ▶ Based on the answers, the token's vector is shifted.

Deep Learning: Stacking Layers

A single attention + MLP block is not enough. Transformers stack these blocks deeply.

- ▶ **Depth:** 12 layers (GPT-2 Small) to 96 layers (GPT-3).
- ▶ Each layer refines the vectors in the sequence, encoding increasingly abstract features:
 - ▶ **Early layers:** Focus on syntax and basic grammar.
 - ▶ **Middle layers:** Focus on local semantic relationships and references.
 - ▶ **Late layers:** Synthesize global context to prepare the prediction.
- ▶ Computation consists of matrix multiplications transforming the vector representations at each step.

Deep Learning Premise: Weights vs. Data

To understand the math, we must separate the model's structure into two categories:

The Brains: Weights

- ▶ Tunable parameters (knobs/dials).
- ▶ Organized in massive **matrices**.
- ▶ Fixed after training.
- ▶ E.g., GPT-3 has 175 billion weights.

The Flow: Data / Tensors

- ▶ Arrays representing current inputs.
- ▶ Pass through the network, constantly changing.
- ▶ E.g., token vectors, intermediate activations.

Summary: Lecture 1 Recap

- ▶ GPTs generate text by repeatedly predicting the next token.
- ▶ Text is tokenized, then represented as coordinate vectors in a high-dimensional space.
- ▶ ****Attention Blocks**** allow tokens to share contextual information.
- ▶ ****MLP Blocks**** process each token's vector independently in parallel.
- ▶ Stacking these operations creates a deep network capable of capturing nuanced meaning.